

Isuru Wijesiri

I develop production AI systems at WSO2

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Education

University of Moratuwa, Computer Science and Engineering

Sri Lanka

- Dean's List: 7 out of 8 semesters (Semesters 1, 3, 4, 5, 6, 7, 8)
- Thesis: Privacy-Preserving Machine Learning on Distributed Graphs (Grade: A+)
- University Award for Best R&D Project (highest marks achieved)

2017 – 2021

Experience

WSO2, Associate Technical Lead (AI R&D)

Sri Lanka

Technical lead for AI capabilities in the WSO2 Integrator platform. Research focus: LLM failures in low-resource code generation and benchmarking LLM agents on enterprise integration.

Feb 2026 – present

8 months

- Designing multi-agent systems and developing/training custom models for agentic workflows.
- IntegrationBench: Leading a 100-task, behavior-graded benchmark that evaluates LLM agents by the runtime behavior of the integrations they build (targeting ICLR 2027).
- Domain-specific SLM: Developing a lightweight small language model for on-prem domain-specific code generation.
- Owning architecture, execution, and delivery of agentic developer workflows, and production readiness (security, scalability, reliability).
- Mentoring engineers and interns to ship high-quality AI features.

WSO2, Senior Software Engineer (AI R&D)

Sri Lanka

Led the effort to bring AI into WSO2 Micro Integrator.

Jan 2024 – Feb 2026

2 years 2 months

- WSO2 Integrator Copilot: Owned from ideation to production. NL to Synapse DSL code generation with version-aware retrieval, XSD/Schematron validation, LSP diagnostics, and automated repair loops. 700+ active daily users.
- WSO2 Integrator Copilot Agent Mode: Cursor/Claude Code-style agentic integration engineer inside VS Code with 23 tools across 8 categories, multi-step planning, and autonomous task execution.
- Integrated domain-specific embedding models into WSO2 Integrator Copilot retrieval system for low-latency semantic code search.
- Optimized operational cost via advanced prompt caching (~90% cost reduction) and production hardening/observability hooks.
- Small Language Model initiative: Initiated domain-specialized SLM training for Ballerina to enable low-latency on-prem code generation.

WSO2, Software Engineer

Sri Lanka

Built open-source products for API Management and Integration.

Jan 2022 – Dec 2023

2 years

- Delivered multiple end-to-end features for WSO2 API Manager and contributed 40+ product improvements.
- Distributed Transaction Counter: Scalable transaction counting for high-throughput API gateways handling 10,000+ TPS with sub-millisecond latency.
- GenAI / Multi-agent framework: Led a low-code framework supporting multiple LLM backends + vector stores, reducing delivery time and standardizing deployments.

University of Moratuwa , Lead Research Collaborator Supervisors: Dr. Surangika Ranathunga (Massey University, NZ) and Dr. Nisansa de Silva (UOM) - Leading Google-funded research on Automatic Post-Editing (APE) for low-resource languages (Sinhala, Tamil). - Proposing controllable minimal-edit post-editing with hidden-state anchoring and dynamic edit-distance constrained decoding. - Driving experimentation, evaluation, and research direction for practical APE workflows.	Sri Lanka Aug 2025 – present 1 year 2 months
Persistent Systems , Research Engineer Developed architectures and POCs for a new API Marketplace intended to host and manage thousands of APIs.	Sri Lanka Mar 2021 – Dec 2021 10 months
University of Moratuwa , Graduate Researcher (Thesis Continuation) Supervisors: Prof. Sanath Jayasena, Dr. Miyuru Dayarathna - Extended undergraduate thesis on privacy-preserving distributed graph ML. - Developed novel common-attribute and common-entity aggregation mechanisms for heterogeneous multi-organization graph learning while maintaining privacy. - Implemented prototype on JasmineGraph distributed graph database.	Sri Lanka June 2021 – Dec 2021 7 months
University of Moratuwa , Undergraduate Thesis Researcher Supervisors: Prof. Sanath Jayasena, Dr. Miyuru Dayarathna - Researched privacy-preserving distributed graph ML and federated GNNs for large-scale link prediction. - Built memory-efficient distributed/federated GCN training enabling training on very large graphs on commodity hardware (JasmineGraph). - Peer-reviewed publication at IEEE Big Data 2020.	Sri Lanka Jan 2020 – June 2021 1 year 6 months
Zeptolytics , Machine Learning Research Intern - Developed end-to-end automated data analytics and Business Intelligence pipeline for the ZEPTO platform. - Resulted in 40% improvement in decision-making speed for client companies. - Built using Python, Scikit-learn, pandas, NumPy. Deployed with Docker and AWS.	Colombo, Sri Lanka June 2019 – Dec 2019 7 months

Publications

Confident but Wrong: A Constrained Decoding Diagnostic for Low-Resource Automatic Post-Editing Isuru Wijesiri github.com/IsuruMaduranga/confident-but-wrong (Findings of EMNLP 2026 (first author))	Jan 2026
EnSiTa: A Trilingual Multi-Domain Parallel Dataset and Benchmark for Domain-Specific Machine Translation Isuru Wijesiri doi.org/10.13140/RG.2.2.28180.87688 (Preprint (targeting TACL))	Jan 2026
IntegrationBench: Evaluating LLM Agents on Enterprise Integration Tasks Isuru Wijesiri (Manuscript in preparation (targeting ICLR 2027, first author))	Jan 2026
Memory-Efficient Graph Convolutional Network-Based Distributed Link Prediction Isuru Wijesiri, Miyuru Dayarathna, Sanath Jayasena ieeexplore.ieee.org/document/9377874 (2020 IEEE International Conference on Big Data (Big Data))	Jan 2020

Anomaly Detection using Deep Reconstruction and Forecasting for Autonomous Systems Isuru Wijesiri, Sachith Gunasekara, Tharindu Wickramasinghe arxiv.org/abs/2006.14556 (arXiv:2006.14556)	Jan 2020
High Accuracy Link Prediction in a Collaborative Multi-Organization Graph Learning System Isuru Wijesiri (Manuscript in preparation)	Jan 2026

Projects

One Code - Cross-Model Agent Harness An agent harness that brings Claude Code-style features to any model and provider.	Jan 2026
WSO2 Integrator Copilot - Agentic Coding Assistant for WSO2 MI - Cursor/Claude Code-style coding agent for enterprise integration with 700+ daily active users. 23 tools across 8 categories: project management, connector management, XML LSP validation, builds, and runtime workflows. +29 percentage-point improvement on HumanEval over base LLM (0-shot) using multi-agent iterative refinement and execution-guided verification.	2023 - present
Lightweight Code Retrieval Models (22M/33M parameters) - Fine-tuned MiniLM-based sentence-transformer models (512-dim) for domain-specific code retrieval using contrastive learning. 97% Recall@10 and 95% MRR@10. Released on Hugging Face (700+ downloads/month).	Jan 2025
Toolflow - Structured Agent Framework Lightweight research framework for LLM-based agents with structured output guarantees and automatic tool invocation.	Jan 2025
Naturalpy - Natural Programming Interface for Python Natural-language-driven programming abstractions for Python with LLM-backed function invocation and code synthesis.	Jan 2025
Distributed Transaction Counter for WSO2 Synapse Scalable transaction counting for high-throughput API gateways handling 10,000+ TPS with sub-millisecond latency.	Jan 2022
Privacy-Preserving ML on Federated Graph Database Systems - Federated learning for collaborative, privacy-preserving graph learning on JasmineGraph distributed graph database. - IEEE Big Data 2020 publication. Technologies: C++, Python, StellarGraph, TensorFlow.	Jan 2020
Anomaly Detection for Autonomous Systems - Self-supervised deep reconstruction and forecasting for drones (video + IMU). 94% accuracy / 0.95 F1 on video, 100% accuracy / 0.98 F1 on IMU. IEEE Signal Processing Cup 2020 runner-up.	Jan 2020

Awards

IEEE Signal Processing Cup 2020 - First Runner-Up - First Runner-Up at ICASSP 2020 finals (team Icarus Inhibition). - Achieved the highest score in the selection round for anomaly detection in autonomous drone systems using multimodal sensor data.	Jan 2020
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University Award for Best R&D Project Highest marks achieved in undergraduate thesis research on distributed graph ML.	Jan 2021
Virtusa Award - Awarded to the student who earned the highest marks for the semester 5 Software Engineering Project. - Issued by Department of CSE, University of Moratuwa.	Jan 2019
Flu Shot Learning Challenge - 18th/4811 (top 0.4%) DrivenData machine learning competition.	Jan 2020
SLIIT Codefest Datathon - Champions Winning team (XGCoders) of nationwide inter-university data mining competition.	Jan 2019
iHack 5.0 - Winners 24-hour inter-university software development hackathon at University of Colombo. - Team ODIGOS designed and developed a mobile/web app solving Sri Lanka's tourism issues. First place out of 59 teams.	Jan 2019

Teaching

University of Moratuwa , Lab Assistant: Programming Fundamentals (CS1033)	Sri Lanka 2020 – 2021
Private Tutoring , Private Tutor (Data Structures, ML, Physics & Maths) Mentored 60+ students.	2017 – 2021
WSO2 Technology Conference 2025 , Conference Presenter Technical deep-dive on blockchain architecture, consensus algorithms, and distributed ledger mechanisms.	Jan 2025
Harness Engineering 101 (Blog Series) , Author A 16-chapter blog series on building LLM agents and harnesses from first principles.	Jan 2026
WSO2 , Mentor & Technical Lead, Integration AI Team Supervise and mentor junior AI engineers and interns.	2024 – present